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Probabity & Statistics Assignment 2

Date:

Q 3.1a

$X = 0, 1, 2$

$P(X=0) = P(1-0.4) \cdot P(1-0.3) = (0.6)(0.7) = 0.42$

$P(X=1) = P(File1^c \cap File2^{NC}) + P(File1^{NC} \cap File2^c)$

$= \cancel{P}(0.4 \cdot 0.7) + (0.6(0.3))$

$= 0.28 + 0.18$

$= 0.46$

$P(X=2) = P(File1^c) \cdot P(File2^c)$

$= (0.4)(0.3)$

$= 0.12$

$\therefore$ pmf $p(x) \Rightarrow$ $P(X=0) = 0.42$

$P(X=1) = 0.46$

$P(X=2) = 0.12$

b) For CDF

so $F(x) = P(X \leq x)$

$F(x \neq 0) = P(X < 0) = 0$

$F(x=1) = P(X<1) = P(X=0) = 0.42$

$F(x=2) = P(X<2) = P(X=0) + P(X=1) = 0.42 + 0.46 = 0.88$

$F(X=3) = P(X<3) = P(X=0) + P(X=1) + P(X=2) =$

$= 0.42 + 0.46 + 0.12$

$= 1$

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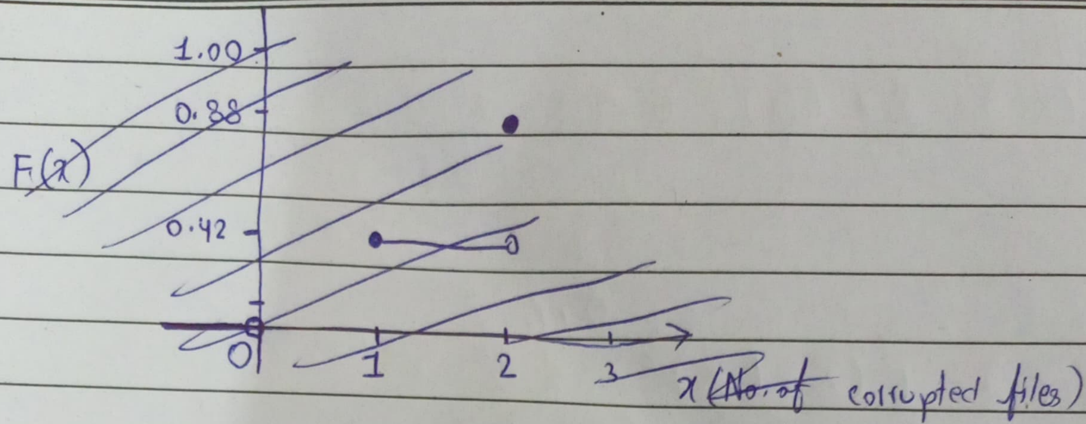

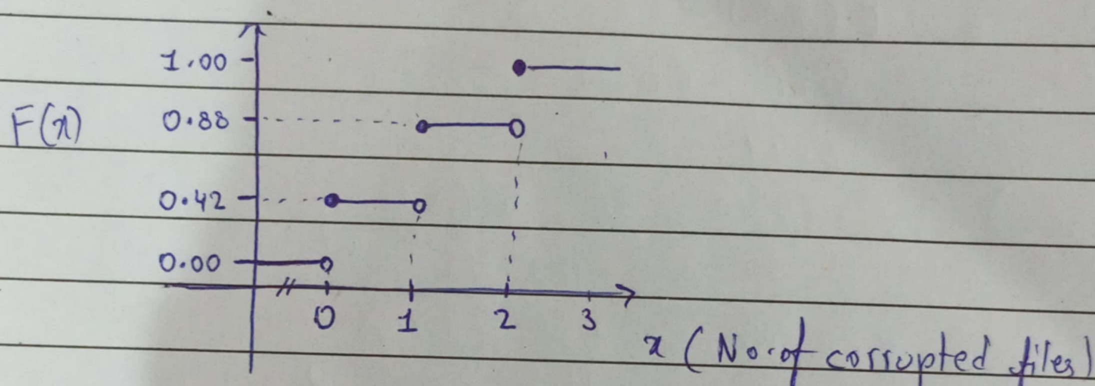

Q 3.2

each blackout cost = \$500 loss

let X = no. of blackout

Total loss = $500X$

$\mu = E(X) = \Sigma x P(x)$

$= 0(P(0)) + 1P(1) + 2(P(2))$ ~~3(P(3))~~

$= 0(0.7) + 1(0.2) + 2(0.1)$

$= 0 + 0.2 + 0.2$

$= 0.4$

Expected loss $= 500 \times 0.4 = 200$ } Ans

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for Variance

$E(x^2) = \sum x^2 P(x) = 0^2 P(0) + 1^2 P(1) + 2^2 (P(2))$

$= 0^2 (0.7) + 1(0.2) + 4(0.1)$

$= 0 + 0.2 + 0.4$

$= 0.6$

$Var(x) = E(X - EX)^2 = \sum (x - u)^2 P(x)$

$= E(x^2) - (E(x))^2$

$= 0.6 - (0.4)^2$

$= 0.44$ Ans

$Var(Loss) = Var(aX) = a^2 Var(x)$

$= 500^2 Var(x)$

$= 500^2 (0.44)$

$= 110,000$ Ans

Variance of daily loss is \$ 110 000 and expected daily loss due to network blackout is \$200

Q 3.3

Data

Total Blocks = 5

Blocks with error = 1

Choose Block = 3

Let X = no. of errors in selected blocks

$E(X) = \frac{n \cdot K}{N}$

n = sample size (aka 3 select box)

K = no. of items with error

N = Total population

$= \frac{3(1)}{5}$

$= 0.6$

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$$Var(X) = E(X) \cdot \frac{N-K}{N} \times \frac{N-n}{N-1}$$

$$= 0.6\left(\frac{5-1}{5} \times \frac{5-3}{5-1}\right)$$

$$= 0.6\left(\frac{4}{5} \times \frac{2}{4}\right)$$

$$= 0.6(0.4)$$

$$= 0.24$$

Q 3.4

for $E(X)$ fair die so each side probability is $\frac{1}{6}$

$$E(X) = \sum_{i=1}^{6} x_i \cdot P(x_i)$$

$$= 1P(1) + 2(P(2)) + 3(P(3)) + 4(P(4)) + 5(P(5)) + 6(P(6))$$

$$= 1(1/6) + 2(1/6) + 3(1/6) + 4(1/6) + 5(1/6) + 6(1/6)$$

$$= 1/6 + 2/6 + 3/6 + 4/6 + 5/6 + 6/6$$

$$= 21/6$$

$$= 3.5$$ Ans

for $Var(X)$

$$Var(X) = E(X^2) - [E(X)]^2$$

for $E(X^2) = 1^2(1/6) + 2^2(1/6) + 3^2(1/6) + 4^2(1/6) + 5^2(1/6) + 6^2(1/6)$

$$= (1+4+9+16+25+36)/6$$

$$= 91/6$$

$$= 15.16$$

$$Var(X) = 15.16 - (3.5)^2$$

$$= 2.91$$ Ans

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Q3.5

Data

Total Program = 12

Upgraded = 5

Selected = 4

a) Probability at least 2 upgraded $P(X \geq 2)$

$P(X \geq 2) = 1 - P(X=0) - P(X=1)$ } equivalent expression

Total way to choose 4 program from 12 $\Rightarrow \binom{12}{4} = \frac{12!}{4!\,(12-4)!} = 495$

$$P(X=0) = \frac{{}^5C_0 \cdot {}^7C_4}{{}^{12}C_4} = 0.0707$$

$$P(X=1) = \frac{{}^5C_1 \cdot {}^7C_3}{{}^{12}C_4} = 0.3535$$

$\therefore P(X \geq 2) = 1 - 0.0707 - 0.3535$

$= 0.5758$ Ans

b) $E(x) = n \cdot \frac{K}{N}$

n = selected program

N = Total Population

K = upgraded

$= 4\left(\frac{5}{12}\right)$

$= 1.6667$ Ans

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Q 3.6

Data:

Total block = 6

Selected = 2

Error = 1

$$E(X) = \frac{nK}{N} = \frac{2(1)}{6} = \frac{1}{3} = 0.333$$ Ans

OR

$$P(X=0) = \frac{{}^5C_2}{{}^6C_2} = 0.666$$ } 0 errors

$P(X=1) = 1 - P(X=0) = 0.333$ } 1 error found

so $E(X) = 0(P(0)) + 1(P(1))$

$= 0(0.666) + 1(0.333)$

$= 0.333$

Q 3.7

Let X_1 = Run in Game 1

X_2 = Run in Gam 2 so $Y = X_1 + X_2$

$E(X_1) = E(X_2) = \sum x P(x)$

$= 0(P(0)) + 1(P(1)) + 2(P(2))$

$= 0(0.4) + 1(0.4) + 2(0.2)$

$= 0.4 + 0.4$

$= 0.8$

$E(Y) = \cancel{E(X_1+X_2)} = E(X_1) + E(X_2) = 0.8 + 0.8 = 1.6$

Ans

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for Variance

$$Var(Y) = Var(X_1) + Var(X_2)$$

$$Var(X_1) = Var(X_2) = E(X^2) - (E(X))^2$$

for

$$E(X^2) = x^2 P x = 0^2(P(0)) + 1^2(P(1)) + 2^2(P(2))$$
$$= 0(0.4) + 1(0.4) + 4(0.2)$$
$$= 0.4 + 0.8$$
$$= 1.2$$

$$Var(X_1) = 1.2 - (0.8)^2$$
$$= 0.56$$

$$\therefore Var(Y) = Var(X_1) + Var(X_2)$$
$$= 0.56 + 0.56$$
$$= 1.12 \quad \text{Ans}$$

Q3.8) $X = 0, 1, 2, 3, 4$ no. of tries it took to find correct password.

$$P(X=0) = 1/4$$
$$P(X=1) = \frac{1}{3}\left(\frac{3}{4}\right) = \frac{1}{4}$$
$$P(X=2) = \frac{3}{4} \times \frac{2}{3} \times \frac{1}{2} = 1/4$$
$$P(X=3) = 1 - \left(\frac{1}{4} + \frac{1}{4} + \frac{1}{4}\right) = 1/4$$

$$\therefore E(X) = 1\left(\frac{1}{4}\right) + 2\left(\frac{1}{4}\right) + 3\left(\frac{1}{4}\right) + 0\left(\frac{1}{4}\right)$$
$$= (1+2+3)/4$$
$$= 6/4$$
$$= 1.5$$

Date:

for Variance

$$E(x^2) = 0^2\left(\frac{1}{4}\right) + 1^2(1/4) + 2^2\left(\frac{1}{4}\right) + 3^2\left(\frac{1}{4}\right)$$
$$= (1+4+9)/4$$
$$= 14/4$$

$$Var(X) = E(X)^2 - (E(x))^2$$
$$= 14/4 - (6/4)^2$$
$$= (14 - \cancel{36}^{9})/4$$
$$= 1.25 \quad Ans$$

Q3.10

F = friday T = Thursday

$$P(F > T)$$

valid case

$$\left.\begin{array}{l} F=1,\ T=0 \\ F=2,\ T=0 \\ F=2,\ T=1 \end{array}\right\}\text{no. of acident}$$

Sol

$$P(1,0) = 0.2(0.6) = 0.12$$
$$P(2,0) = 0.2(0.6) = 0.12$$
$$P(2,1) = 0.2(0.2) = 0.04$$

$$\therefore P(F > T) = 0.12 + 0.12 + 0.04$$
$$= 0.28$$

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Q 3.11

a) Joint PMF of (X, Y)

$X = \min \quad Y = \max$

- if $x < y$: 2 outcomes $\rightarrow (x, y)$ and (y, x)
- if $x = y$: 1 outcome

$$P(X = x, Y = y) = \begin{cases} \frac{2}{36} & x < y \\ \frac{1}{36} & x = y \end{cases} \qquad x > y \quad 0$$

X \ Y	1	2	3	4	5	6
1	1/36	2/36	2/36	2/36	2/36	2/36
2	0	1/36	2/36	2/36	2/36	2/36
3	0	0	1/36	2/36	2/36	2/36
4	0	0	0	1/36	2/36	2/36
5	0	0	0	0	1/36	2/36
6	0	0	0	0	0	1/36

b) $P(X=1, Y=1) = 1/36$

$P(X=1) = 1/36$ and $P(Y=1) = 1/36$

$P(X=1) \cdot P(Y=1) = 1/36 \cdot 1/36 = 1/1296$

$\therefore$

$P(X=x, Y=y) \neq P(X=x) \, P(Y=y)$

Hence they are not independent

c) $X = 1$ ~~(6 outcome)~~

$X = 2$

$X = 3$

$X = 4$

$X = 5$

$X = 6$

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$P(X=1) = 1+2+2+2+2+2 = \frac{11}{36}$
$P(X=2) = 0+1+2+2+2+2 = 9/36$
$P(X=3) = 0+0+1+2+2+2 = 7/36$
$P(X=4) = 0+0+0+1+2+2 = 5/36$
$P(X=5) = 0+0+0+0+1+2 = 3/36$
$P(X=6) = 0+0+0+0+0+1 = 1/36$

d) $P(X=2, Y=5) \Rightarrow P(Y=5 \mid X=2) = \frac{P(Y=5, X=2)}{P(X=2)}$

$= \frac{2/36}{9/36}$

$= \frac{2}{9}$

$= 0.222$ Ans

Q 3.12)
a) $P(X=0) = 0.5 + 0.2 = 0.7$ $\quad$ $P(X=1) = 0.2 + 0.1 = 0.3$
$P(Y=0) = 0.5 + 0.2 = 0.7$ $\quad$ $P(Y=1) = 0.2 + 0.1 = 0.3$

$P(X=0, Y=0) = 0.5$
for $x=0, y=0 \Rightarrow P(X=0) \cdot P(Y=0) = 0.7 \times 0.7 = 0.49$

since $0.5 \neq 0.49$ the condition fails
hence X and Y are not independent

b) For $X+Y$

(X, Y)	A: $X+Y$	B: $X-Y$	Probab
$(0,0)$	0	0	0.5
$(0,1)$	1	-1	0.2
$(1,0)$	1	1	0.2
$(1,1)$	2	0	0.1

Let $A = X+Y$

$B = X-Y$

$P(A=1, B=1) = 0.2$

but

~~$P(A=1) = P(X=1) \cdot P(Y=0) = 0.2(0.2) = 0.4$~~

~~$P(B=1) = P(X=1$~~

$P(A=0) = X+Y=0 \Rightarrow 0.5$

$P(A=1) = P(0,1) + P(1,0) = 0.2 + 0.2 = 0.4$

$P(A=2) = P(1,1) = 0.1$

$P(B=0) = P(0,0) + P(1,1) = 0.5 + 0.1 = 0.6$

$P(B=1) = P(1,0) = 0.2$

$P(B=-1) = P(0,1) = 0.2$

now as from table $P(A=1, B=1) = 0.2$

but $P(A=1) \cdot P(B=1) = 0.4(0.2) = 0.08$

They are not equal hence they are not independent

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Q 3.13

X\Y	0	1	2	P(x)
0	0.2	0	0.3	0.5
1	0	0.1	0	0.1
2	0.3	0	0.1	0.4
P(y)	0.5	0.1	0.4	

a) $Z = X + Y$

X	Y	Z = X + Y	Probab
0	0	0	0.2
0	2	2	0.7
1	1	2	0.7
2	0	2	0.7
2	2	4	0.4

$\therefore$

Z	P(Z)
0	0.2
2	0.7
4	0.1

when $X+Y=2 \Rightarrow P(X=1, Y=1) + P(X=0, Y=2) + P(X=2, Y=0)$

$= 0.1 + 0.3 + 0.3$

$= 0.7$

$X+Y=4 \Rightarrow P(X=2, Y=2)$

$= \cancel{0.4}\ 0.1$

b) $U = X - Y$

X	Y	U = X - Y	Probab
0	0	0	0.4
0	2	-2	0.3
1	1	0	0.2
2	0	2	0.3
2	2	0	

$P(U=-2) = P(X=0, Y=2) = 0.3$

$P(U=2) = P(X=2, Y=0) = 0.3$

$P(U=0) = P(X=1, Y=1) + P(X=0, Y=0) + P(X=2, Y=2)$

$0.2 + 0.1 + 0.1 = 0.4$

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$\therefore$

U	$P(U)$
0	$0.2+0.1+0.1 = 0.4$
-2	0.3
2	0.3

c) $V = XY$

X	Y	$V = XY$	Probab
0	0	0	0.8
0	2	0	0.8
1	1	1	0.1
2	0	0	0.8
2	2	4	0.1

$P(V=0) = P(X=0, Y=0) + P(X=0, Y=2) + P(X=2, Y=0)$
$= 0.2+0.3+0.3 = 0.8$

$P(V=1) = P(X=1, Y=1)$
$= 0.1$

$P(V=4) = P(X=2, Y=2)$
$= 0.1$

$\therefore$

V	$P(V)$
0	0.8
1	0.1
4	0.1

Q 3.14

X	Y	$Z = XY$	P
1	2	2	0.1
1	3	3	0.4
2	1	2	0.06
2	2	4	0.1
2	3	6	0.1
3	1	3	0.06
3	2	6	0.04
4	1	4	0.1
4	2	8	0.04

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so

Z	Probab(Z)
2	0.16
3	0.46
4	0.20
6	0.14
8	0.04

$P(Z=2) = P(X=2, Y=1) + P(X=1, Y=2)$

$0.06 + 0.10 = 0.16$

$P(Z=3) = P(X=1, Y=3) + P(X=3, Y=1)$

$0.40 + 0.06$

0.46

$P(Z=4) = P(X=2, Y=2) + P(X=4, Y=1)$

$0.1 + 0.1 = 0.2$

$P(Z=6) = P(X=2, Y=3) + P(X=3, Y=2)$

$0.1 + 0.04$

$P(Z=8) = P(X=4, Y=2)$

$= 0.04$

Q3.15

a) ~~$P(X \geq 1) = 1 - P(X=0)$~~ $P(\text{at least one}) = 1 - P(X=0, Y=0)$

~~$= P$~~

$= 1 - 0.52$

$= 0.48$ Ans

b) $P(X=0) = 0.52 + 0.14 + 0.06 = 0.72$

$P(Y=0) = 0.52 + 0.20 + 0.04 = 0.76$

so $P(X=0) \cdot P(Y=0) = (0.72)(0.76) = 0.5472$

$P(X=0, Y=0) = 0.52$

As $P(X=0, Y=0) \neq P(X=0) \cdot P(Y=0)$

$0.52 \neq 0.5472$

$\therefore$ it isn't dependent

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3.16)

a)

(X,Y)	P
$(0,0)$	0.6
$(0,1)$	0.1
$(1,0)$	0.1
$(1,1)$	0.2

$P(0,0) = 0.6$

$P(X=0) = 0.6+0.1 = 0.7$

$P(Y=0) = 0.6+0.1 = 0.7$

$P(X=0) \cdot P(Y=0)$

$= (0.7)(0.7)$

$= 0.49$

$P(Y=1) = 0.1+0.2 = 0.3$

$P(X=1) = 0.1+0.2 = 0.3$

$P(0,0) \neq P(X=0)P(Y=0)$

As $0.6 \neq 0.49$

$\therefore$ it's not independent

b) $E(X+Y)$

sol

$E(X+Y) = E(X) + E(Y)$

$= 0.3 + 0.3$

$= 0.6$

for $E(X)$

$E(X) = 0(0.7) + 1(0.3)$

$= 0.3$

for $E(Y) = 0(0.7) + 1(0.3)$

$= 0.3$

Q3.17)

a) Each Share (A or B)

+1 profit with Prob 0.5

-1 profit with Prob 0.5

and they are independent

Risk = variance

- High var is more unpredictable (risky)
- low var is more stable (less risky)

$E(X) = 1(0.5) + (-1)(0.5) = 0$

on average no gain no loss.

$Var(X) = E(X^2) + (E(X))^2$

$= (0.5 + 0.5) + (0)^2 = 1$

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each share has variance = 1

a) Var (total) = n Var (one share)

= 100 (1)

= 100

b) 50A + 10B

Var = 50 (1) + 10 (1)

= 60

c) 40A + 12B

Var = 40 (1) + 12 (1)

= 52

The lowest rick is of Portfolio C (40A + 12B)

Q 3.18) for X

$E(X) = -3(0.3) + 0(0.2) + 3(0.5)$

$= -0.9 + 1.5$

$= 0.6\%$

for Y

$E(Y) = -3(0.4) + 3(0.6)$

$-1.2 + 1.8$

0.6%

for X ⇒ $E(X^2) = 9(0.3) + 0 + 9(0.5) = 2.7 + 4.5 = 7.2$

for Y ⇒ $E(Y)^2 = 9(0.4) + 9(0.6) = 9$

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$Var(X) = E(X^2) - (E(X))^2 = 7.2 - 0.6^2 = 6.84$

$Var(Y) = E(Y^2) - (E(Y))^2 = 9 - 0.6^2 = 8.64$

a) $\text{Profit} = \frac{\text{investment (return)}}{100}$

$\underset{(Z)}{\text{Profit}} = \frac{1000\,X}{100} = 10X$

$E(Z) = 10E(X) = 10(0.6) = 6$

$Var(Z) = 10^2\,Var(X) = 100(6.84) = 684$

b) $50A + 10B$

$\text{Profit} = \frac{(50\times10)\,X}{100} + \frac{10(50)\,Y}{100}$

$= 5X + 5Y$

$E(5X+5Y) = 5E(X) + 5E(Y) = 5(0.6) + 5(0.6) = 6$

$Var(5X+5Y) = 5^2 Var(X) + 5^2 Var(Y) = 25(6.84) + 25(8.64)$

$= 387$

c) 20 shares of B

$\underset{(Z)}{\text{Profit}} = \frac{1000\,Y}{100} = 10Y$

$E(Z) = 10(0.6) = 6$

$Var(Z) = 10^2(8.64) = 864$

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The least risky one is B as it has the least variance.
The most risky one is C as it has the highest variance,

Though all portfolio give expected profit.

Q3.19)

For Share A

$$E(X) = 2(0.5) + (-2)(0.5) = 0$$
$$E(X^2) = 2^2(0.5) + 2^2(0.5) = 4$$
$$Var(X) = E(X^2) - (E(X))^2$$
$$= 4 - 0$$
$$= 4.$$

for Share B

$$E(Y) = 4(0.2) + (-1)(0.8) = 0$$
$$E(Y^2) = 4^2(0.2) + 1(0.8) = 4$$
$$Var(Y) = E(Y^2) - (E(Y))^2$$
$$= 4 - 0$$
$$= 4$$

a) 100 shares of A

$Var = 100^2(4) = 40000$ drs

$E(100X) = 100\,E(X) = 100(0)$
$= 0$ drs

b) 100 shares of B

$Var = 100^2(4) = 40000$ drs

$E(100Y) = 100\,E(Y) = 100(0) = 0$ drs

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c) $50A + 50B$

$\cancel{50E(A) + 50E}(B)$ $E(50A+50B) = 50E(x) + 50E(y)$

$\cancel{50}$ $= 50(0) + 50(0)$

$= 0$ Ans

$Var(50A+50B) = 50^2 Var(x) + 50^2 Var B$

$= 2500(4) + 2500(4)$

$= 20000$ Ans

Q3.20) a) $X \sim$ Binomial where $n = 20$ $p = 0.05$ $q = 0.95$

$P(X=3) = {}^{20}C_3 (0.05)^3 (0.95)^{17}$

$= 0.0596$

b) not included

Q3.21) $X \sim$ Binomial $(n=20, p=0.4)$

$P(X \geq 10) = 1 - P(X \leq 9)$

so $P(X \leq 9) = \sum_{k=0}^{9} \binom{20}{k} (0.4)^k (0.6)^{20-k}$

$= 0.755337$

$P(X \geq 10) = 1 - 0.755337$

$= 0.2446$

Q3.22) $X \sim$ Binomial $(n=16, p=0.05)$

$P(X>3) = 1 - P(X \leq 3)$

so $P(X \leq 3) = \sum_{k=0}^{16} \binom{16}{k} (0.05)^k (0.95)^{16-k}$

$= 0.99299 \approx 0.992$

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$P(X>3) = 1-0.992\cancel{9}$ or $1-0.99298$

$= 0.008\cancel{0}$ $= 0.0070$

Q 3.23a) $X \sim \text{Binomial}(15, 0.05)$

$P(X \geq 4) = 1 - P(X \leq 3)$

$P(X \leq 3) = \sum_{\cancel{x}=0}^{3} \binom{15}{k} (0.05)^{x} (0.95)^{15-x}$

$= 0.994$

$P(X \geq 4) = 1 - 0.994$

$= 0.006$

b) not included

Q 3.24a) $X \sim \text{Binomial}(10, 0.2)$

$P(X \geq 5) = 1 - P(X \leq 4)$

$P(X \leq 4) = \sum_{K=0}^{4} \binom{10}{K} (0.2)^{K} (0.8)^{10-K}$

$= 0.9672$

$P(X \geq 5) = 1 - 0.9672$

$= 0.0328$

24b) Not included.